



Curvature-Based Stealth Design and Radar Analysis

End Semester Poster Presentation

Submitted by

Ayush Saw (2023meb1331) , Baljinder Kaur (2023meb1334), Jasnoor Kaur (2023meb1351) and Kondala Seershika (2023meb1355)

Supervisor : Prof. Harpreet Singh Co-Supervisor : Dr. Satwinder Jit Singh



Introduction

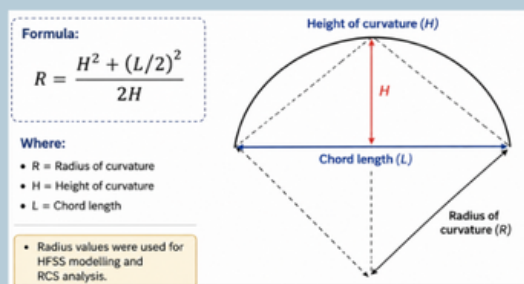
- There are several classes of stealth techniques, such as camouflage, acoustic-based, infrared-based, radar-absorbing material (RAM)-based, and geometry-based methods, through which the stealthiness of a structure can be achieved. Among these, geometrical shaping reduces direct electromagnetic reflections by redistributing scattered waves in different directions
- In this work, the effect of surface curvature on electromagnetic scattering and Radar Cross Section (RCS) was analyzed using HFSS simulation and experimental VNA measurements for flat and curved aluminum structures

Problem Statement

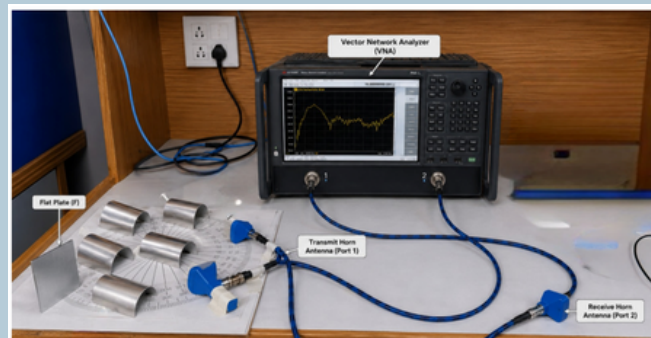
Flat metallic surfaces strongly reflect incident electromagnetic waves back toward the radar source, resulting in high Radar Cross Section (RCS) and increased radar detectability. There is a need to investigate how surface curvature can redistribute electromagnetic reflections and improve stealth performance without relying only on complex stealth materials or coatings.

Methodology

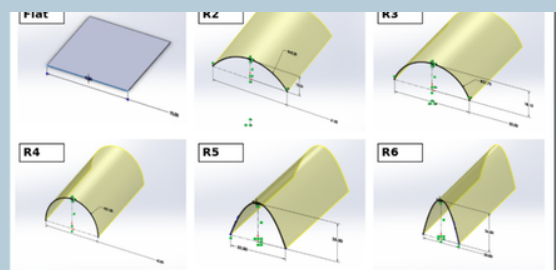
1. Radius Calculation



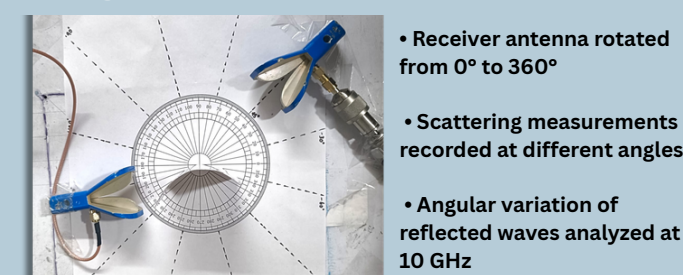
4. Experimental Setup(VNA)



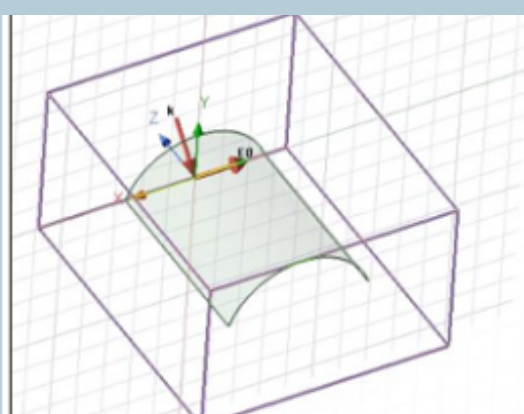
2. Curvature Design



5. Angular Measurement



3. HFSS Simulation



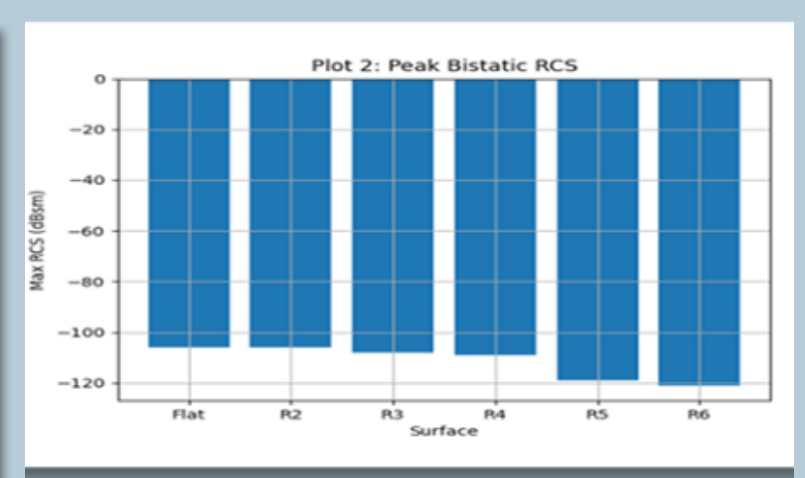
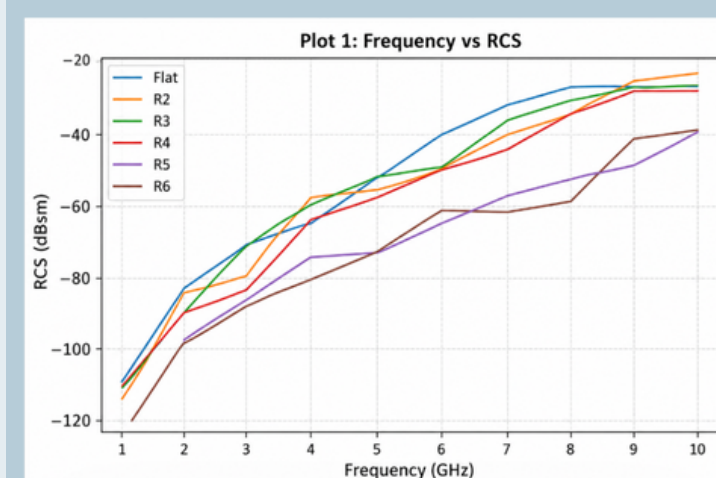
6. Comparative Analysis

- HFSS bistatic RCS compared with experimental S21 results
- Curved geometries showed reduced directional reflection
- R5 geometry exhibited smoother scattering distribution and improved stealth performance

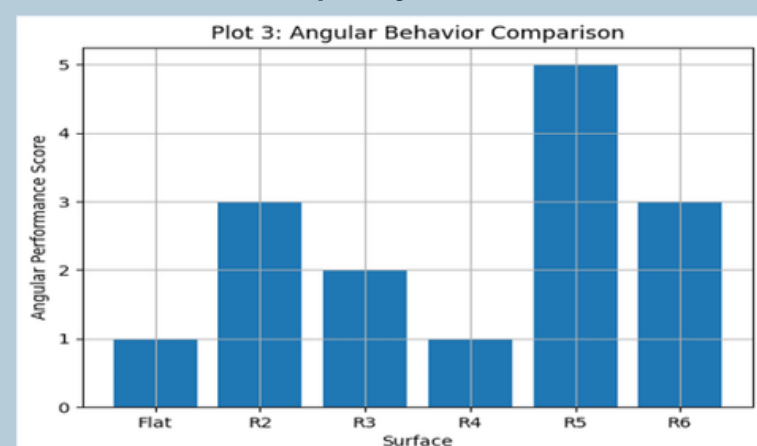
Project Objectives

- To design and analyze flat and curved aluminum geometries for stealth applications.
- To study the influence of surface curvature on electromagnetic wave scattering and Radar Cross Section (RCS).
- To identify the optimum curvature configuration that minimizes electromagnetic reflection using ANSYS HFSS simulations.
- To experimentally validate the scattering behavior using VNA-based S21 measurements and compare with simulation results.

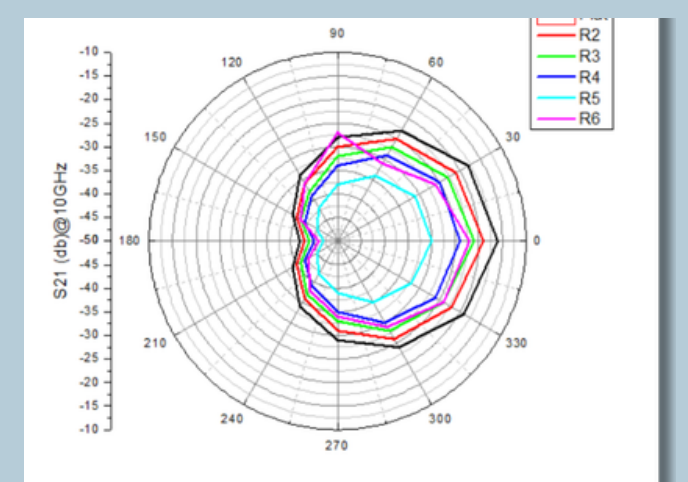
Results



Plot 1: Frequency vs RCS

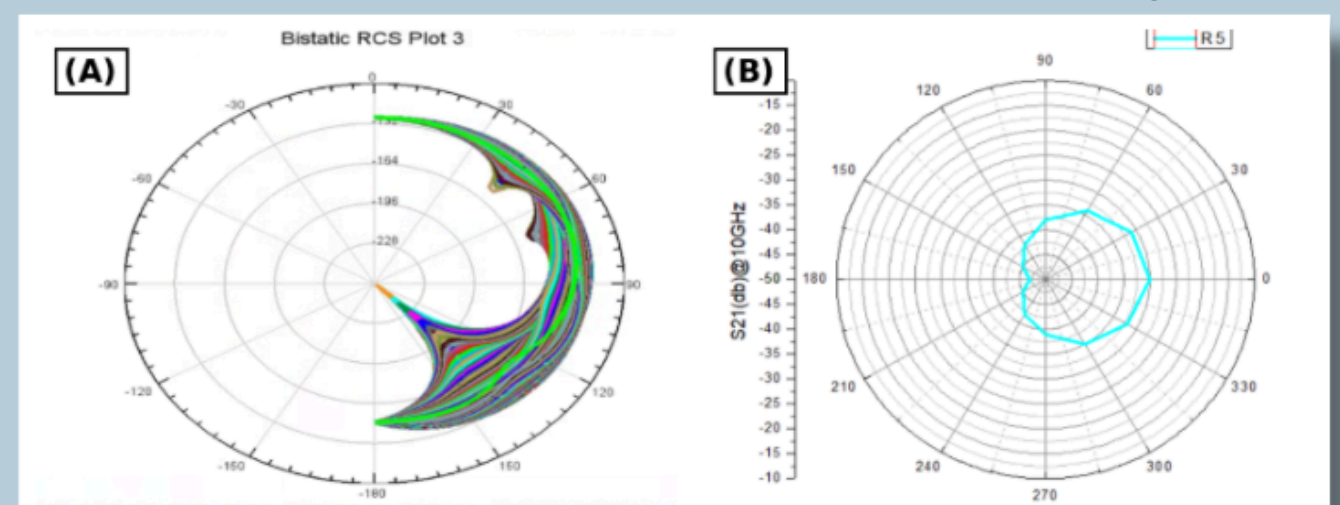


Plot 2: Peak Bistatic RCS



Plot 3: Angular Behavior Comparison

Plot 4: Polar Scattering Distribution



(A) HFSS Bistatic RCS Polar Plot

(B) Experimental S21 Polar Plot

- HFSS bistatic RCS and experimental S21 plots showed similar angular scattering behavior with maximum scattering near 170°-180°.
- Both analyses confirmed that the R5 geometry provided smoother scattering distribution and improved stealth performance.

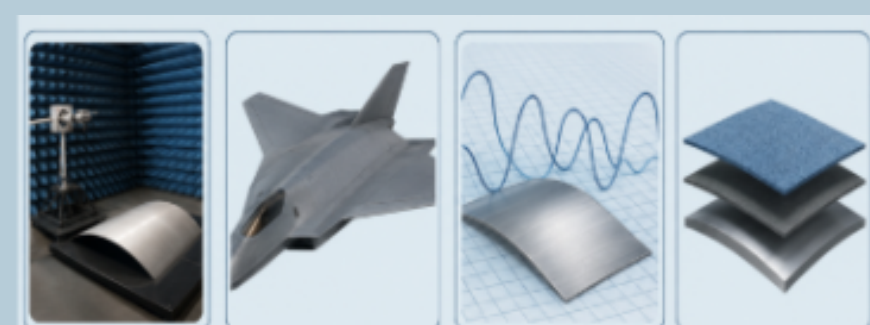
Conclusions

- Curvature-based stealth analysis was successfully performed using ANSYS HFSS and experimental VNA measurements.
- Curved geometries redistributed electromagnetic waves and reduced strong directional reflections compared to the flat plate.
- HFSS bistatic RCS and experimental S21 analyses showed similar scattering trends at 10 GHz.
- Among all geometries, R5 exhibited the lowest reflection, smoother scattering distribution, and best stealth performance.
- The study confirmed that optimized surface curvature effectively improves electromagnetic stealth characteristics.

References

- E. F. Knott, J. F. Shaeffer, and M. T. Tuley, Radar Cross Section, SciTech Publishing, 2004.
- C. A. Balanis, Advanced Engineering Electromagnetics, Wiley, 1989.
- M. U. Ullah et al., "A progression in the techniques of reducing RCS for stealth applications," Alexandria Engineering Journal, 2024.

Future Scope



- RCS Measurement
- Full-Scale Extension
- Multi-Band Analysis
- Advanced Materials

Acknowledgements

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